KI-SL: An Efficient and Secure Knowledge Index Based Swarm Learning Method of Data Collaboration

Haiyan Kang, Chong Zhang

Abstract—**The rapid development of IoV plays a pivotal role in the construction of efficient intelligent transportation systems. However, its heterogeneous data collaboration process faces multiple challenges, such as insufficient collaboration efficiency, low communication efficiency, and difficulties in data privacy protection. To address these issues, this study proposes an efficient and secure Knowledge Index based Swarm Learning method (KI-SL) of data collaboration. Firstly, we construct a swarm learning framework based on blockchain identity authentication. Secondly, a Knowledge Distillation based on Knowledge Index algorithm is proposed, construct knowledge index via hash encoding to enable cross-node knowledge sharing, establishing association relationships by combining local adjacency lists with nearest-neighbor search, and designing an asynchronous distillation mechanism to realize dynamic knowledge updates. Additionally, an adaptive sample selection and prototype optimization strategy is introduced, and secure broadcast of the knowledge index is realized through blockchain, thereby improving model performance and reducing communication overhead. Furthermore, a Quantization Differential Privacy algorithm is presented, which applies stochastic quantization to the refined gradients and superimposes Laplacian noise to strengthen data privacy protection. Finally, experimental validation was conducted on real public datasets. The results demonstrate that the proposed method achieves maximum model accuracies of 89.21% and 92.78%, respectively. It significantly improves communication efficiency while ensuring the security of data collaboration, and provides a new method for data collaboration in distributed environments.**

Index Terms—**Data Collaboration, Swarm Learning, Blockchain, Knowledge Index, Knowledge Distillation, Privacy Preservation.**

I. Introduction

Amid the accelerated transition of the global transportation system toward intelligence and automation, Intelligent Transportation Systems (ITS) have become a critical infrastructure driving the transformation of future mobility. As a core supporting technology within ITS, the Internet of Vehicles (IoV) enables multi-dimensional interactions, including Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), Vehicle-to-Pedestrian (V2P), and Vehicle-to-Network (V2N), thereby reconstructing the operational mode of transportation networks and providing data and computing support for autonomous driving, collaborative perception, and real-time traffic dispatching [1],[2],[3]. In recent years, with the continuous evolution of 6G communication and edge intelligence, the IoV system is evolving towards ultra-low latency, high reliability, and intelligent autonomy, and its operating environment exhibits higher dynamics, heterogeneity, and distributed characteristics [4],[5],[6],[7],[8],[9].

However, this trend also gives rise to unprecedented challenges. Firstly, massive volumes of data in the IoV are generated in real time by highly distributed vehicle nodes; such data not only exhibit significant Non-Independent and Identically Distributed (Non-IID) characteristics, but also possess strong spatiotemporal correlation and dynamism [2],[6]. Under such complex data conditions, traditional collaborative learning methods tend to suffer from issues such as unstable model convergence and degraded generalization ability [10], [11]. Secondly, privacy and security concerns have become increasingly severe. Once vehicle trajectories, driving behaviors, and location information are exposed, they may lead to severe personal privacy exposure, social security threats, and even economic losses [4],[12]. In an open, dynamic, and potentially adversarial network environment, traditional centralized data processing architectures can no longer meet the security and scalability requirements of large-scale IoV applications, revealing prominent issues such as computational bottlenecks, single-point failures, and privacy leakage risks [5],[6].

To address the aforementioned limitations, the distributed collaborative learning has gradually emerged and become a crucial direction for the intelligent evolution of the IoV. In recent years, the academic community has proposed a variety of approaches to achieve cross-node data collaboration and privacy protection through distributed training [13],[14],[15],[16]. However, existing methods still have significant shortcomings in highly dynamic IoV scenarios [17]: (1) Highly dynamic network topologies lead to a significant increase in communication costs and system instability, which restricts the timeliness and robustness of model training [14],[15]; (2) Non-IID data distributions and heterogeneous computing resources make it difficult to balance collaboration efficiency and model performance [1],[16]. More critically, how to realize a traceable, secure, and efficient data sharing and learning mechanism remains a scientific challenge that

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Haiyan Kang is with School of Computer Science, Beijing Information Science and Technology University, Beijing 102206, China (E-mail: kanghaiyan@126.com).

Chong Zhang is with School of Computer Science, Beijing Information Science and Technology University, Beijing 102206, China (E-mail: zgx2543@163.com).

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urgently requires breakthroughs [18],[19],[20]. Therefore, the current field of collaborative learning still faces the following core scientific issues that urgently need to be addressed: (1) In the highly dynamic and non-IID IoV environment, how to achieve stable and efficient cross-node data collaboration; (2) Under the conditions of limited communication resources and continuous expansion of node scale, how to significantly reduce communication overhead while maintaining model performance; (3) In scenarios without a global trusted third party, how to achieve secure and privacy friendly data collaborative learning.

In response to the aforementioned issues, we have conducted in depth research, with the primary contributions summarized as follows:

1) We propose an efficient and secure Knowledge Index based Swarm Learning method (KI-SL) of data collaboration, establishing a comprehensive learning framework from the node level to the global level. By employing a knowledge index mechanism, the method drives a knowledge distillation algorithm for efficient knowledge sharing among heterogeneous nodes, thereby optimizing the model and reducing communication overhead. In addition, a quantization differential privacy algorithm is designed to effectively enhance the protection of sensitive data.

2) We propose a Knowledge Distillation Based on Knowledge Index (KDKI) algorithm. It constructs a knowledge index via hash encoding to enable knowledge sharing among nodes, and combines local adjacency lists with efficient nearest-neighbor search to build cross-node association relationships. Meanwhile, it designs an asynchronous distillation mechanism, which allows nodes to receive knowledge representations from neighboring nodes based on similar features for dynamic optimization, and enables secure broadcast through blockchain. Furthermore, it introduces adaptive sample selection and prototype feature optimization strategies to achieve dynamic updates of the knowledge index, improve model performance and reduce communication overhead.

3) We propose and design a Quantization Differential Privacy (QDP) algorithm that combines stochastic quantization with differential privacy. This algorithm converts gradient data into discrete values via stochastic quantization and adds Laplacian noise, thereby effectively preventing the leakage of sensitive information and bolstering data privacy safeguards.

4) Experimental results on real public datasets verify the effectiveness of the proposed method, indicating that KI-SL outperforms existing methods in terms of model performance, communication efficiency, and privacy preservation.

The rest of this paper is organized as follows: Section II discusses the related research on knowledge distillation, differential privacy, swarm learning (SL), and data collaboration in IoV environments; Section III elaborates on the framework of KI-SL, as well as the algorithm design and analysis of KDKI and QDP; Section IV presents the experimental settings and evaluation results; Section V conducts ablation study; finally, the conclusion section summarizes this study and discusses future research directions.

II. Related Work

Efficient and secure data transmission and communication have always been research hotspots in the field of distributed collaborative learning. This section will sequentially review research progress from three major directions: knowledge distillation, differential privacy, and swarm learning, and briefly elaborate on the challenges of data collaboration in vehicle networking environments.

A. Research progress of knowledge distillation

In recent years, attention has increasingly turned to the integration of knowledge distillation with distributed machine learning to bolster the security of data transmission. Wu et al. [21] proposed the FedKD framework, which employs an adaptive mutual-distillation strategy alongside dynamic gradient compression to substantially reduce communication overhead, while enabling efficient collaborative training under privacy-preserving constraints. Shao et al. [22] proposed Selective-FD, a framework that uses a selective knowledge-sharing mechanism to filter out misleading or ambiguous information, thereby enhancing both the performance and robustness of federated distillation in heterogeneous environments. Wu et al. [23] developed FedCache, which incorporates knowledge caching and privacy-preserving sample-hashing techniques to achieve highly efficient personalized model distillation, improving communication efficiency and ensuring model consistency—an innovative solution for data collaboration in edge-intelligent systems. Building on this foundation, Pan et al. [24] presented FedCache2.0, which combines federated data distillation with a device-centric cache-sampling strategy to further optimize model efficiency and accuracy in scenarios lacking public data, demonstrating significant promise across heterogeneous devices and dynamic network conditions.

B. Research progress of differential privacy

With the ever-increasing demand for privacy protection, distributed machine learning methods that integrate differential privacy have garnered widespread attention. Wu et al. [25] proposed a federated learning framework combining adaptive gradient descent with differential privacy, which dynamically adjusts the learning rate to optimize training efficiency while resisting background-knowledge attacks, thereby achieving a balance between privacy protection and utility in communication-constrained environments. Kang et al. [26] proposed a data compliance utilization method based on adaptive differential privacy and federated Learning, ADP-BCFL, which utilizes blockchain to achieve secure storage and efficient querying of user data summaries. Meanwhile, it proposes an adaptive differential privacy mechanism, combining with the ADAM algorithm, adaptively adjusts the parameter clipping threshold according to the characteristics of gradient parameters, controls the amount of noise injection, and ensures the accuracy of the global model while resisting inference attacks. Lyu et al. [27] presented a secure and efficient federated learning framework grounded in stochastic quantization; by incorporating a low-complexity quantization module that converts gradients into discrete values and injecting noise for privacy, it substantially reduces communication overhead while ensuring robust privacy protection, offering a novel solution for resource constrained scenarios.

C. Research and Application of Swarm Learning

As a machine-learning framework that integrates blockchain and federated learning, Swarm Learning [28] enables decentralized collaborative model training to effectively address the data-privacy and cooperation challenges inherent in centralized learning, demonstrating immense application value in domains such as healthcare and intelligent transportation. Building on this paradigm, Saldanha et al. [29] explored the use of Swarm Learning for medical image analysis, leveraging blockchain to realize fully decentralized AI model training and thereby resolving both privacy-protection and cross-institutional collaboration issues in medical data sharing. Furthermore, Wang et al. [3] introduced the IoV-SFDL framework, which incorporates a trust-weight prediction algorithm to assess the efficacy and robustness of self-learning group relative to the global model; IoV-SFDL outperforms traditional approaches in model accuracy, convergence speed, and communication-link overhead. Finally, Kang et al. [30] proposed the Swarm Mutual Learning (SML) framework, which employs reciprocal distillation between local and proxy models to enhance accuracy, with its effectiveness validated on real-world datasets.

D. Challenges of data collaboration in IoV

With the continuous breakthroughs in communication and computing technologies, especially the in-depth application of the Internet of Things (IoT) and Edge Computing (EC), the development of the Internet of Vehicles (IoV) has been strongly supported, laying a solid technical foundation for its large-scale expansion [31],[32]. Against this backdrop, the volume of data generated by traffic participants has experienced exponential growth; particularly in intelligent IoV scenarios such as autonomous driving, these large-scale data not only undertake core perception and decision-making tasks but also play a crucial role in the safety and performance of intelligent transportation [33]. However, in the IoV environment, especially under conditions of dense nodes and frequent data interactions, data collaboration still faces two core challenges: *1) High communication overhead:* Vehicle nodes in the IoV need to frequently share model features, gradient updates, and training results. Under the conditions of a large number of nodes and frequent training iterations, direct transmission of complete model parameters or a large number of prototype features will not only lead to a sharp increase in communication load [34], [35], but also significantly restrict the efficiency of collaborative learning and the scalability of the system [36], [37]. *2) Data Privacy Protection:* Local training data of vehicle nodes usually contains sensitive information such as driving behavior patterns, trajectory and location [38],[39],[40],[41]. Direct cross-node transmission without encryption or compression can easily give rise to risks of privacy leakage and data abuse [42]. Therefore, how to achieve effective data privacy protection while ensuring collaboration efficiency has become a core issue that urgently needs to be addressed in IoV collaborative learning.

Although collaborative learning frameworks have demonstrated certain effectiveness in scenarios such as healthcare and the Industrial Internet of Things (IIoT) in recent years, in the complex environment of the Internet of Vehicles (IoV), which is characterized by high dynamism, node heterogeneity, and high-frequency communication [43], existing methods generally suffer from drawbacks such as insufficient adaptability, limited training efficiency, and weak privacy protection. These issues seriously restrict their practical application in intelligent transportation and autonomous driving.

III. KI-SL Design

An efficient and secure Knowledge Index based Swarm Learning method (KI-SL) is proposed to address the communication efficiency and data privacy challenges arising of data collaboration. The overall framework is shown in Fig. 1, consists of three layers: user, permissioned blockchain, and collaborative swarm network. At the user layer, each vehicle node trains on its local data and, upon completion, uploads an encrypted model increment together with its knowledge index. The blockchain guarantees tamper proof logging and security for all model updates. The collaborative swarm network orchestrates knowledge interaction and collaborative training among vehicle nodes. Through this fully decentralized collaboration mechanism, KI-SL supports asynchronous operations and dynamic adjustment, thereby ensuring uninterrupted model convergence and effective data exchange.

The core workflow of KI-SL is as follows. Firstly, each vehicle node trains a local model on its private data, generates model features, encodes them via a hash function to produce a knowledge index, and broadcasts this knowledge index through a smart contract. Upon training completion, nodes upload the hash digests and knowledge indexes to the collaborative swarm network via the blockchain, updates its local model by retrieving similar-feature indices published by other nodes, and simultaneously integrate adaptive sample selection to achieve efficient utilization of distilled data. Secondly, the blockchain provides immutable data records and security guarantees, while smart contracts automate the verification of incoming model updates. Third, prior to uploading, each node applies gradient normalization and random quantization to its model increment, reducing the volume of data transmitted. Then, the quantized gradients preserve privacy by adding noise controlled by the privacy budget ε. Finally, the collaborative swarm network aggregates all received gradients to form a global model, which is then disseminated back to each node for local model refinement. KI-SL encompasses two key algorithms, the Knowledge

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Distillation Based on Knowledge Index (KDKI) algorithm and the Quantization Differential Privacy (QDP) algorithm, whose design principles and execution flows are detailed in Sections 3.1 and 3.2, respectively, and the description of the main symbols involved is provided in TABLE I.

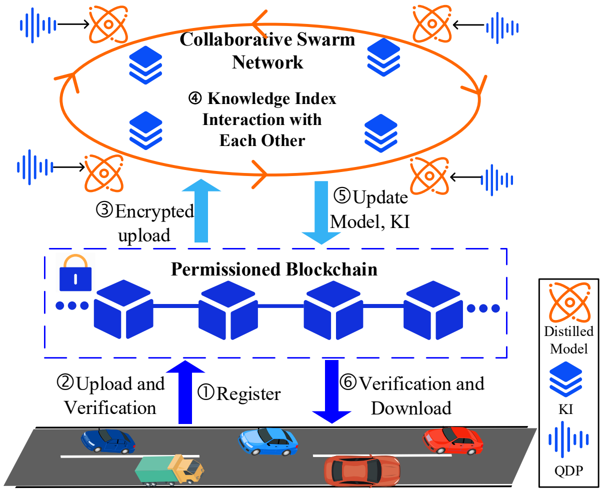

Fig. 1. Framework of KI-SL.

TABLE I
Description of Main Symbols

Symbol	Description
$KI(\cdot)$	Knowledge Index
$F_c(\cdot)$	The deep pre trained encoder
$r_{u,v}$	The low dimensional hash mapping corresponding to $a_{u,v}$
$S(u,v)$	Local hash table
$T(b)$	Label Index Table for Mapping Relationships
$K(u,v)$	Knowledge vector associated with (u,v)
$LA(u,v)$	Local adjacency list of (u,v)
R	The number of neighboring samples in KI
$F_u(\cdot)$	Local feature extraction network of u
$z_{u,v}$	Feature embedding representation of $a_{u,v}$
$\bar{z}_{u,v}$	Integrated knowledge representation
τ	Adaptive Sampling Parameter
C	Gradient clipping threshold
ΔG	ℓ_2-sensitivity of the gradient
ε	Differential privacy budget

A. Knowledge Distillation Based on Knowledge Index (KDKI)

To optimize the model and mitigate the communication overhead incurred by parameters transmission, we propose the Knowledge Distillation Based on Knowledge Index (KDKI) algorithm, which achieves efficient and secure cross node collaborative learning by indexing knowledge representations instead of directly sharing raw data. The workflow of which is illustrated in Fig. 2. Its core principles are as follows. Firstly, each node locally utilizes a deep encoder to perform hash mapping on data, generates low-dimensional digests, and constructs knowledge indexes, the knowledge index is used to store information and interact with other nodes. Subsequently, nodes securely exchange hash digests and intermediate knowledge representations through blockchain, and build cross-node association relationships by means of HNSW nearest-neighbor search and local adjacency lists, retrieving similar knowledge from neighboring nodes based on local sample features for dynamic updates. Next, during the distillation optimization phase, nodes improve the effectiveness of cross-node knowledge sharing through prototype feature mapping and kernel ridge regression loss functions, while integrating adaptive sample selection strategies to achieve efficient utilization of distilled data. Finally, in each training round, nodes iteratively update model parameters based on integrated knowledge, and securely broadcast the updated knowledge indexes to the swarm network for asynchronous optimization. The initialization and training(updating) procedures for the knowledge index are detailed in Fig. 3.

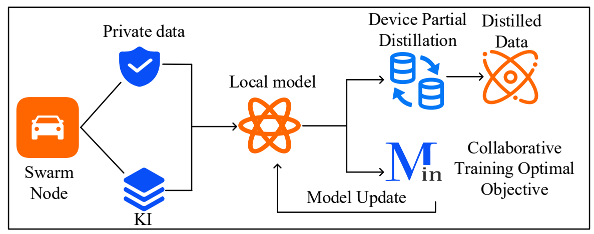

Fig. 2. Flow chart of KDKI.

The specific steps of the algorithm are as follows:

Step1 Knowledge Index Initialization and Hash Encoding. In the swarm network, each node $u \in \{1, 2, \ldots, U\}$ extracts feature representations using its local dataset $Q_u = \{(a_{u,v}, b_{u,v})\}_{v=1}^{N_u}$. The node generates a hash value $r_{u,v}$ (a low dimensional learnable embedding representation) for each sample $a_{u,v}$ through a dedicated deep pre-trained encoder F_c.

$$r_{u,v} = F_c(a_{u,v}). \tag{1}$$

Where N_u is the number of samples in Q_u, and $a_{u,v}, b_{u,v}$ are the v-th data and label in Q_u respectively. The hash values generated by each node are securely broadcast to neighboring nodes through smart contract along with the sample indices (u, v). Since F_c is a deep neural network, it transforms the original data into low-dimensional hash values through multiple non-linear mappings. Therefore, nodes only share the hash values without exposing the original data.

The generated hash value $r_{u,v}$ is stored in the local hash table $S(u,v)$. Meanwhile, the label table $T(b_{u,v})$ is updated, and the knowledge vector $K(u,v)$ is initialized as a zero vector. Where d represents the number of classes.

$$S(u,v) = r_{u,v}\,, \tag{2}$$

$$T(b_{u,v}) = T(b_{u,v}) \cup \{(u,v)\}\,, \tag{3}$$

$$K(u,v) = \vec{0}_d. \tag{4}$$

Step2 Swarm Network Interaction and Relationship Establishment. After the knowledge index initialization is completed, each node continuously receives the hash values $r_{u,v}$ of other nodes through the smart contract, and updates its label table T, knowledge vector K and local hash table S. This process continues until all nodes have collected the sample hash values of other nodes in the network. Subsequently, nodes establish the association relationships between sample

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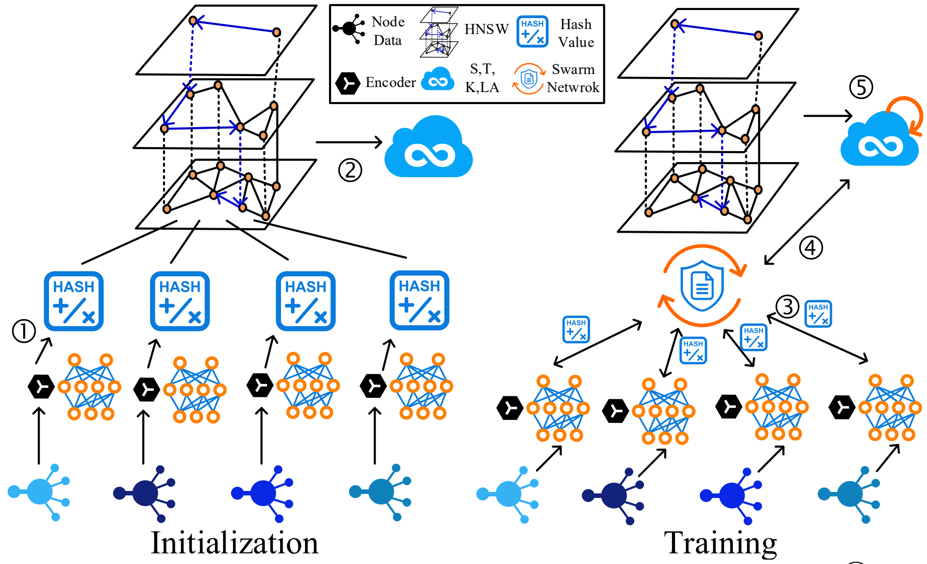

Fig. 3. Knowledge index initialization and training/updating phase execution process. ①Hash encode and upload. ②Knowledge index initialization. ③Swarm network interaction. ④Knowledge acquisition and asynchronous optimization. ⑤Knowledge dynamic update.

indices through the efficient Hierarchical Navigable Small World (HNSW) nearest-neighbor search algorithm. This process uses the similarity of sample hash values to match the most relevant samples.

$$\max_{(j_1,k_1),\ldots,(j_R,k_R)} \sum_{s=1}^{R} sim\left(S(u,v), S(j_s,k_s)\right). \quad (5)$$

Meanwhile, the results of the association relationships are stored in the local adjacency table $LA(u,v)$, which records the hash values of the R most relevant neighbor samples to this sample.

$$LA(u,v) \leftarrow \{(j_1,k_1),\ldots,(j_R,k_R)\}. \quad (6)$$

Step3 Dynamic Knowledge Update and Asynchronous Optimization. During the model optimization process, each node first extracts the knowledge representation $z_{u,v}$ of the local sample $(a_{u,v}, b_{u,v})$ through the deep neural network F_u.

$$z_{u,v} = F_u(a_{u,v}). \quad (7)$$

Notable, the knowledge distillation allows nodes to asynchronously acquire relevant knowledge from other nodes to optimize local models. Based on the pre-established local adjacency list $LA(u,v)$, a node retrieves the most relevant knowledge to the sample index (u,v) from the knowledge index KI, and updates its local knowledge vector $K(u,v)$.

$$KI(u,v;r_{u,v}) = K(LA(u,v)). \quad (8)$$

Each node sends the generated knowledge representation $z_{u,v}$ along with the sample index (u,v) to other nodes in the swarm network via the smart contract. Subsequently, the node retrieves R pieces of knowledge related to the sample (u,v) from the local knowledge index $KI(u,v;r_{u,v})$ in accordance with Equation (8).

$$\left(z_{u,v}^{(1)}\right),\left(z_{u,v}^{(2)}\right),\ldots,\left(z_{u,v}^{(R)}\right) = KI(u,v;r_{u,v}). \quad (9)$$

Among them, $\left(z_{u,v}^{(s)}\right)$ is the s-th piece of knowledge obtained for the given sample index (u,v).

After that, $\left(z_{u,v}^{(s)}\right)$ and the sample index (u,v) are securely sent to neighboring nodes in the swarm network. Each node securely shares these intermediate knowledge representations through the blockchain and downloads the averaged and integrated knowledge representation $\bar{z}_{u,v}$ from the knowledge index of neighboring nodes for further model training.

$$\bar{z}_{u,v} = \frac{1}{R}\sum_{s=1}^{R}\left(z_{u,v}^{(s)}\right). \quad (10)$$

Then, these intermediate knowledge representations are securely shared via the blockchain, ensuring that each node can download the updated knowledge representation $\bar{z}_{u,v}$ from neighboring nodes for further local model training.

Meanwhile, the node continuously updates its local knowledge index to ensure that the latest knowledge state is obtained each time. For a newly uploaded sample index (u,v), the node acquires and returns the corresponding knowledge based on the stored local adjacency list LA and knowledge vector mapping K.

$$K(u,v) \leftarrow z_{u,v}\,. \quad (11)$$

Step4 Distillation optimization. During the knowledge distillation process, effective cross-node knowledge sharing is achieved through the knowledge index, ensuring that each node can obtain distilled data from neighboring nodes to optimize its local model. When performing swarm knowledge distillation, a node decomposes its local model into a feature extractor Ψ_u and a classifier φ_u, and acquires the distilled knowledge of neighboring nodes through the knowledge index.

For each sample $(a_{u,v}, b_{u,v})$, the feature extractor outputs the feature representation $z_{u,v} = \Psi_u(a_{u,v})$ according to the semantic similarity between samples, and then the classifier outputs the prediction result $\varphi_u(z_{u,v})$ .These feature representations are securely broadcast to other nodes through the smart contract. The distilled data of each node is indexed and managed by its unique identifier, so as to achieve efficient knowledge update in the knowledge index KI and initialize the distilled prototypes on the node devices.

$$KI[node,u] \leftarrow \hat{Q}_u. \quad (12)$$

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Where $\hat{Q}_u$ represents the synthetic data extracted at node u.

During the distillation process, each node initializes the prototypes in two ways. One is to select a local sample for each class during the initial communication. The other is to receive the distilled data from other nodes in subsequent communications, where a random replacement function σ controls whether to update the current prototype with the received data. The generated distilled data is stored in the knowledge index.

$$P_u = \begin{cases} KI[node, \sigma(u)], & KI[node, \sigma(u)] \neq \emptyset \\ Q_u^{init}, & otherwise \end{cases}. \quad (13)$$

Here, P_u represents the prototype samples set that needs to be further optimized into synthetic data. And Q_u^{init} is a subset selected from Q_u, which satisfies the following constraint conditions:

$$b \neq b' \vee b = b' \wedge a \neq a', \forall (a, b) \wedge (a', b') \in Q_u^{init}. \quad (14)$$

During the distillation process, nodes calculate the similarity between the prototype feature mapping and the local data feature mapping through the Gram matrix. For the prototype $(a_p, b_p) \in Q_u^{init}$ of node u in class b_p ,the formula for calculating the distance between it and the local sample features is:

$$G_{u,v,p} = \Psi_u(a_{u,v}) \cdot \left(\Psi_u(a_p)\right)^T. \quad (15)$$

The Gram matrix of the prototype features is:

$$G_p = \Psi_u(a_p) \cdot \left(\Psi_u(a_p)\right)^T. \quad (16)$$

This similarity is used to measure the effectiveness of knowledge sharing and serves as the optimization basis in the knowledge distillation process, where feature embeddings are used as inputs to the loss function. The knowledge distillation objective L_p is optimized through the kernel ridge regression loss function.

$$\min_{a_p} L_p = \min_{a_p} \frac{1}{2} \left\| b_p - G_{u,v,p}\left(G_p + \lambda I\right)^{-1} b_{u,v} \right\|^2. \quad (17)$$

Here, I is the identity matrix, and λ is the regularization strength. Through this optimization process, nodes continuously update their models and absorb knowledge from other nodes to improve the generalization ability and performance of the models.

After obtaining the distilled data on node u, it is updated and stored in the local knowledge index KI, so that the latest distilled knowledge can be continuously accessed and utilized in subsequent communication rounds.

$$KI[node, u] \leftarrow P_u. \quad (18)$$

Step5 Optimization objectives. To improve local model performance and enhance subsequent distillation effects, nodes periodically interact with other nodes in the swarm network, requesting distilled data from the index of neighboring nodes for optimization. The optimization objective of co-training is as follows:

$$\min_{\theta_u} L_{p,train} = \sum_{(a_{u,v}, b_{u,v}) \in Q_u} L_{CE}\left(z_{u,v}, b_{u,v}; \theta_u\right) + \xi \left(\sum_{(a',b') \in Sel_u} L_{CE}\left(z_{u,v}', b'; \theta_u\right) \right). \quad (19)$$

The knowledge index samples and allocates storable knowledge based on the node label frequency p_c^u forming an adaptive sample selection strategy Sel_u.

$$Sel_u = \bigcup_{d=1}^{d} RS(KI[d], (\tau + (1-\tau) \cdot p_c^u) \cdot |KI[d]|). \quad (20)$$

Where τ is a parameter used to balance model performance and communication overhead. As τ increases, the sampling ratio also increases, prompting more knowledge sample exchange between nodes. However, this may also incur higher communication burdens.

In the initial stage, each node u calculates its local label distribution based on the label frequency p_c^u.

$$p_c^u = \frac{|\{(a_{u,v}, b_{u,v}) | (a_{u,v}, b_{u,v}) \in Q_u, b_{u,v} = d\}|}{|Q_u|}. \quad (21)$$

Where L_{CE} denotes the cross-entropy loss, which measures the difference between the model output and true labels. The function ξ determines whether to introduce distilled knowledge from neighboring nodes according to the status of the knowledge index.

$$\xi = \begin{cases} a, & KI[node, u] \neq \emptyset \\ 0, & otherwise \end{cases}. \quad (22)$$

This implies that when distilled data exists in the knowledge index, the node will proactively utilize these data to optimize the local model; otherwise, the node relies solely on local data for training.

Step6 Model update and iteration. In each training round, intermediate knowledge representations $\left(z_{u,v}^{(s)}\right)$ are continuously exchanged among nodes, and relevant integrated knowledge representations $\bar{z}_{u,v}$ are obtained from the knowledge index. Security verification is then conducted via blockchain smart contracts to ensure the reliability of knowledge sources, while the local knowledge index KI is updated to reflect newly acquired knowledge. At the end of each training round, nodes update the model parameters. The update formula for the model is:

$$\theta_u = \theta_u - \eta \cdot \nabla_{\theta_u} \min_{\theta_u} L_{p,train}(\theta_u). \quad (23)$$

Among them, θ_u denotes the model parameters of node u. The learning rate η controls the magnitude of changes in model parameters during each update. $\nabla_{\theta_u} \min_{\theta_u} L_{p,train}(\theta_u)$ represents the gradient of the loss function $\min_{\theta_u} L_{p,train}(\theta_u)$ with respect to the model parameters θ_u. The loss function $\min_{\theta_u} L_{p,train}(\theta_u)$ of node u is used to measure the difference between the model output and true labels. This loss function combines the cross-entropy loss L_{CE} and the KL divergence loss L_{KL} , where the cross-entropy loss L_{CE} measures the difference between the student model and true labels.

$$L_{CE} = -\sum_{v=1}^{N} b_v \log(\hat{p}_v). \quad (24)$$

While L_{KL} measures the difference between the teacher model (i.e., integrated knowledge) and the student model.

$$L_{KL} = \sum_v \bar{z}_{u,v} \log\left(\frac{\bar{z}_{u,v}}{z_{u,v}}\right). \quad (25)$$

By optimizing this objective, the student model gradually absorbs knowledge from the teacher model, thereby improving its performance.

Step7 Dynamic update and broadcast of knowledge index. After training concludes, the knowledge index $KI(u,v)$ is dynamically updated and broadcast to other nodes in the swarm network via smart contracts, ensuring that all nodes can utilize the latest knowledge index for model optimization. During the next training round, nodes continue to extract the most relevant knowledge to their local data based on the new knowledge index, thereby further enhancing model performance. Additionally, the entire distillation and optimization process is asynchronous—nodes can perform local optimization without waiting for synchronous operations across all nodes.

B. Quantization Differential Privacy (QDP)

In order to secure the upload of the distilled model to the swarm network, we propose a Quantization Differential Privacy (QDP) algorithm. Its core principles are as follows. Firstly, the gradients of the distilled model are quantized into discrete values, thereby reducing data precision and substantially lowering transmission volume. Subsequently, during transmission, Laplace noise is added according to the privacy budget, ensuring that even if intercepted, the original information cannot be reconstructed. The specific steps are as follows:

Step1 Gradient calculation. In each round of local model training, the node first performs gradient calculation on the model parameters optimized through distillation. The gradient reflects the trend of change in the loss function with respect to the model parameters and is used to guide the update of model parameters. Let L denote the loss function,θ denote the model parameters, and the input data and labels be x and y, respectively. The gradient G calculation formula is:

$$G = \nabla_{\theta} L(\theta; x, y). \tag{26}$$

Step2 Gradient clipping and normalization. To strictly define gradient sensitivity and satisfy the prerequisite assumptions of the differential privacy mechanism, gradient clipping based on the ℓ_2-norm is first performed prior to quantization. Let the clipping threshold be C; then the gradient clipping operation is defined as:

$$G \leftarrow G \cdot min\left(1, \frac{C}{\|G\|_2}\right). \tag{27}$$

This operation ensures that the ℓ_2-norm of any gradient vector does not exceed the constant C, thereby preventing abnormal gradients from causing excessive sensitivity to the privacy mechanism.

Gradient sensitivity ΔG is defined as the maximum magnitude of change between the corresponding gradient outputs of adjacent datasets (differing by only one sample). Since each gradient vector satisfies $\| G \|_2 \leq C$ after clipping, for any pair of adjacent datasets D and D', there is:

$$\Delta G = \| G(D) - G(D') \|_2 \leq \| G(D) \|_2 + \| G(D') \|_2 \leq 2C. \tag{28}$$

Thus, an explicit upper bound of the gradient sensitivity can be derived:

$$\Delta G \leq 2C. \tag{29}$$

Subsequently, to meet the requirements of subsequent quantization processing, normalization is performed on the gradient tensor G to map it to the interval [0,1], thereby eliminating differences between different scales and parameter dimensions and ensuring the stability and consistency of the quantization process. The normalization formula is:

$$G' = \frac{G - min_{val}}{max_{val} - min_{val} + \omega}. \tag{30}$$

Where min_{val} and max_{val} are the minimum and maximum values of the gradient tensor G, respectively, and ω is an extremely small positive number introduced to prevent the denominator from being zero.

Step3 Stochastic Quantization. The normalized gradient tensor G' is mapped to a discrete value set composed of L quantization levels. The quantization steps are as follows:

Step3-1 Calculation of quantization interval width δ. This value is determined by the quantization levels L.

$$\delta = \frac{1}{L}. \tag{31}$$

Step3-2 Quantization Interval Determination. For the normalized gradient G', compute the lower bound c_m and upper bound c_{m+1} of its quantization interval.

$$c_m = \left\lfloor \frac{G'}{\delta} \right\rfloor \times \delta\,, \tag{32}$$

$$c_{m+1} = c_m + \delta\,. \tag{33}$$

Step3-3 Quantization probability calculation. To ensure the unbiasedness of the quantization process, a stochastic quantization strategy is adopted within the interval $[c_m, c_{m+1}]$, the quantization probability needs to be calculated. The probabilities of quantizing to the lower bound c_m and upper bound c_{m+1} are defined as follows:

$$p_{c_m} = \frac{c_{m+1} - G'}{\delta}. \tag{34}$$

$$p_{c_{m+1}} = \frac{G' - c_m}{\delta}. \tag{35}$$

It is evident that $p_{c_m} + p_{c_{m+1}} = 1$.

Step3-4 Randomly select the quantization value. Based on the calculated probabilities, a quantized value is selected within the interval c_m and c_{m+1} through a random selection method.

$$\tilde{G} = \begin{cases} c_m, & p_{c_m} = \frac{c_{m+1} - G'}{\delta} \\ c_{m+1}, & p_{c_{m+1}} = \frac{G' - c_m}{\delta} \end{cases}. \tag{36}$$

Step4 Add noise. To satisfy ε-differential privacy, a single noise injection operation is performed on the gradient only after quantization is completed. According to the Laplace mechanism, the noise distribution is defined as:

$$Lap(x; 0, f) = \frac{1}{2f} exp\left(-\frac{|x|}{f}\right). \tag{37}$$

Where the noise magnitude f is related to the privacy budget ε and the gradient sensitivity ΔG.

$$f = \frac{\Delta G}{\varepsilon}\,. \tag{38}$$

Combined with the gradient clipping in Step 2, it follows that $\Delta G = 2C$. Finally, noise injection is performed on the quantized gradient, yielding the differentially private gradient $\tilde{G}_{priv}$:

$$\tilde{G}_{priv} = \tilde{G} + n, n \sim Lap\left(0, \frac{2C}{\varepsilon}\right). \tag{39}$$

Step5 Inverse normalization. Remap the gradient with added noise $\tilde{G}_{priv}$ back to the original gradient range.

$$\tilde{G}_{rec} = \tilde{G}_{priv} \cdot (max_{val} - min_{val} + \omega) + min_{val}. \tag{40}$$

Step6 Privacy Budget Accumulation for Multi-round Training. During the processes of multi-round local training and multiple model uploads, privacy loss accumulates with the number of rounds. Let ε_0 be the privacy budget consumed per gradient update round. Under the application of the basic **Sequential Composition** theorem, the total privacy budget satisfies the following equation:

$$\varepsilon_{total} = T \cdot \varepsilon_0. \tag{41}$$

Where T denotes the total number of rounds of local training and uploads.

Step7 Outputs the final gradient. After completing gradient quantization, adding noise, and denormalization. Use the final gradient $\tilde{G}_{rec}$ after quantization and noise injection to update the model parameters.

$$\theta' = \theta - \eta \tilde{G}_{rec}. \tag{42}$$

Where η is the learning rate, θ is the current model parameter, and θ' is the updated model parameter.

C. KI-SL Analysis

We conducted a comprehensive evaluation of the proposed KI-SL along five dimensions: model performance, computational complexity, communication complexity, and security and blockchain overhead.

1) Analysis of Model Performance

To evaluate the effectiveness of KI-SL in Non-IID data scenarios, we first analyze its optimization objective. The objective function of the KDKI algorithm in the local training phase is $\min_{\theta_u} L_{p,train}$.

Theorem 1 Distillation Consistency: Under the condition that the knowledge index $KI(u,v)$ can approximately represent the similarity of cross-node sample distributions, and that the nearest neighbor retrieval error and asynchronous update delay are bounded, the optimal solution θ_u^* of Equation (19) converges to a stable point of the weighted risk minimization solution that incorporates the neighbor knowledge distillation term. That is:

$$\theta_u^* \to R_{local}(\theta_u) + \xi R_{distill}(\theta_u). \tag{43}$$

It ensures the convergence consistency and the upper bound of generalization stability under Non-IID data conditions.

Proof:

Let the local empirical risk be defined as:

$$R_{local}(\theta_u) = \sum_{(a_{u,v}, b_{u,v}) \in Q_u} L_{CE}\big(z_{u,v}, b_{u,v}; \theta_u\big). \tag{44}$$

Let the distillation knowledge risk be defined as:

$$R_{distill}(\theta_u) = \sum_{(a', b') \in Sel_u} L_{CE}\big(z_{u,v}', b'; \theta_u\big). \tag{45}$$

Then the optimization objective can be rewritten as:

$$L_{p,train}(\theta_u) = R_{local}(\theta_u) + \xi R_{distill}(\theta_u). \tag{46}$$

Considering the nearest neighbor retrieval error ϵ_{nn} and the delay error ϵ_{async} introduced by asynchronous distillation, the distillation risk term can be regarded as an empirical risk term with perturbation, whose upper bound of error satisfies:

$$| R_{distill} - \hat{R}_{distill} | \leq O(\epsilon_{nn} + \epsilon_{async}). \tag{47}$$

According to the Empirical Risk Minimization theory, when the sample size satisfies $|Q_u| + |Sel_u| \to \infty$ and the perturbation is bounded, we have:

$$\mathbb{E}[L(\theta_u^*)] - L(\theta^*) \leq O\left(\sqrt{\frac{1}{|Q_u| + |Sel_u|}}\right) + O(\epsilon_{nn} + \epsilon_{async}). \tag{48}$$

Where θ^* denotes the global optimal solution under ideal conditions. This theorem indicates that KDKI can still maintain a controllable upper bound of generalization error under realistic conditions with retrieval errors and asynchronous updates, rather than undermining training stability.

Q.E.D.

2) Analysis of Computational Complexity

The global computational complexity of SL is given by:

$$C_{SL} = O(MNd). \tag{49}$$

Where M denotes the number of global participating nodes, N is the number of data samples, and d is the dimension of model parameters. This indicates that the computational cost of SL exhibits a linear superposition, making it difficult to maintain good computational efficiency in highly heterogeneous scenarios.

By contrast, KI-SL significantly reduces the computational overhead of both individual nodes and the entire system through the KDKI algorithm. The computational complexity of a single node mainly consists of three components: (1) **Hash index update**: Hash encoding is performed on N samples, with a complexity of $O(Nd)$; (2) **Gradient quantization and differential privacy processing**: In each iteration, gradient quantization and noise injection are required, with a complexity of $O(d)$; (3) **Neighbor knowledge sharing**: Each node only needs to process the knowledge representations of R neighbors in the neighbor set $LA(u,v)$, instead of traversing all global nodes, with a complexity of $O(Rd)$ (where $R \ll M$). Thus, the computational complexity of a single node is:

$$C_{KI-SL} = O(Nd + Rd + d) \approx O(Nd + Rd). \tag{50}$$

It is worth noting that KI-SL does not reduce the order of global computational complexity in the worst-case scenario in the asymptotic sense; instead, by restricting cross-node interactions to local neighborhoods, it reduces the scale of nodes actually participating in computation and communication per round from M to R, thereby significantly cutting down system-level constant terms and concurrent scheduling overheads.

When $R \ll M$ and there exists a constant $c \ll N$ such that $R \leq cN$, the global computational complexity of SL is:

$$C_{SL} = O(MNd). \tag{51}$$

The global computational complexity of KI-SL is:

$$\begin{aligned} C_{KI-SL}^{gloabl} &= O(MNd + MRd) < O(MNd + MNd) \\ &= O(2MNd). \end{aligned} \tag{52}$$

Thus, the global computational complexity of KI-SL is much lower than that of SL:

$$C_{KI-SL}^{gloabl} < C_{SL}^{gloabl}. \tag{53}$$

3) Analysis of Communication Complexity

In SL, each node is required to upload complete model parameters during the training process. The communication overhead of per-round model iteration is:

$$Comm_{SL} = O(d). \tag{54}$$

The global communication overhead is:

$$Comm_{SL}^{total} = O(Md). \tag{55}$$

By leveraging the KDKI algorithm and QDP algorithm, KI-SL significantly reduces communication redundancy. Its communication overhead consists of three components: (1) **Hash value transmission**: The hash encoding $r_{u,v}$of each sample is uploaded, with dimension of $h \ll d$, and complexity of $O(Nh)$; (2) **Knowledge representation transmission**: Nodes upload the neighbor knowledge representation $\bar{z}_{u,v}$, with a complexity of $O(Rd)$; (3) **Quantized gradient transmission**: The quantized gradient $\widetilde{D^p}$ is uploaded, with a complexity of $O(d)$. Thus, the communication complexity of a single node for KI-SL is given by:

$$Comm_{KI-SL} = O(Nh + Rd + d). \tag{56}$$

From Equations (54) and (56), taking the ratio gives:

$$\frac{Comm_{KI-SL}}{Comm_{SL}} = O\left(\frac{Nh}{d} + \frac{R}{d} + 1\right). \tag{57}$$

Since the $d \gg R$ in the scenario of distributed large-scale nodes, Equation (57) can be further simplified to:

$$\frac{Comm_{KI-SL}}{Comm_{SL}} = O\left(\frac{Nh}{d} + R + 1\right). \tag{58}$$

When $h \ll d$ and $R \ll M$, this ratio is much smaller than the $O(M)$-order communication complexity of SL's full model upload. Thus, the communication complexity of KI-SL is lower than that of SL, and the compression ratio satisfies:

$$\frac{Comm_{KI-SL}}{Comm_{SL}} = O\left(\frac{Nh}{d} + \frac{R}{M} + 1\right). \tag{59}$$

4) Analysis of Security

KI-SL employs a three-fold mechanism comprising hash knowledge index, quantization differential privacy, and blockchain consensus, each designed to defend against different types of potential attacks. Its **threat models** include: (1) **Reconstruction attack**: An attacker attempts to reverse-engineer the original feature $a_{u,v}$ through intermediate transmission results; (2) **Gradient inference attack**: An attacker infers the distribution of users' private data by collecting or analyzing global gradients; (3) **Tampering attack**: Malicious nodes upload tampered gradients to undermine the global model.

Under the assumption that most nodes are honest, KI-SL satisfies the following security properties to resist the aforementioned three types of attacks:

(1) **Hash Irreversibility**: The sample hash value $r_{u,v}$ is a low dimensional feature index representation generated by the pre-trained encoder $F_c(\cdot)$. Therefore, when the attacker does not know the encoder parameters and can only observe the index output, the original feature reconstruction problem is equivalent to solving an underdetermined inverse problem in the high-dimensional space, whose solution space satisfies:

$$|\left\{a_{u,v} \mid F_c(a_{u,v}) = r_{u,v}\right\}| \gg 1. \tag{60}$$

It follows that:

$$\Pr\left[\, a_{u,v} \mid r_{u,v} \,\right] \ll 1. \tag{61}$$

Thus, given the hash value $r_{u,v}$, the probability that the attacker accurately locates the unique original feature $a_{u,v}$ is extremely low, which effectively defends against reconstruction attacks.

(2) **Quantization Differential Privacy Protection**: During the gradient upload phase, Laplace noise is added to the quantized gradient $\tilde{G}$ thereby satisfying the ε-differential privacy constraint. Specifically, for any pair of adjacent datasets D and D', the following holds:

$$Pr\left[\tilde{G} \mid D\right] \le e^{\varepsilon} Pr\left[\tilde{G} \mid D'\right]. \tag{62}$$

Given that the gradient clipping threshold is C, the ℓ_2-sensitivity of the gradient can be explicitly bounded as:

$$\Delta G \le 2C. \tag{63}$$

Thus, when the noise scale is set to $\frac{2C}{\varepsilon}$, the Quantized Differential Privacy (QDP) mechanism strictly satisfies the standard definition of differential privacy.

Furthermore, in the scenario of multi-round local training and multiple gradient uploads, in accordance with the basic **Sequential Composition** theorem, the overall privacy budget satisfies Equation (41). This demonstrates that the QDP mechanism provides quantifiable and accumulable privacy protection for the gradient upload process within the formal differential privacy framework, thus effectively defending against gradient inference attacks.

(3) **Blockchain Verified Integrity**: The hash chain structure of the blockchain ensures that any single point tampering will disrupt the hash correlation between blocks and be detected promptly; its decentralized redundant storage property requires that an attacker must control a majority of nodes simultaneously to tamper with the global ledger, an act nearly infeasible in large scale collaboration. Meanwhile, Byzantine Fault Tolerance (BFT) consensus mechanisms ensure that updates are only written when nodes exceeding the threshold reach a consensus. Thus, tampering will not take effect when the number of malicious nodes does not meet the fault tolerance ratio.

Based on the aforementioned blockchain properties, the following conclusion can be drawn:

Suppose an attacker attempts to tamper with a certain global update result. To do so, the attacker must simultaneously modify the hash values of this block and all subsequent blocks in the ledger, while maintaining consistency among more than the consensus threshold of nodes across the entire network. Since the hash of each block depends on the hash of the previous block, this means the attack cost grows exponentially with the length of the blockchain.

Under the Byzantine Fault Tolerance (BFT) protocol, as long as the proportion of malicious nodes is below the security threshold, any tampering behavior will be rejected by the majority of honest nodes. Thus, it is proven that blockchain-based global update records theoretically possess strict immutability, which can effectively defend against tampering attacks and ensure the integrity and credibility of the KI-SL training process.

5) Analysis of Blockchain Overhead

Although blockchain provides decentralized trusted verification capability, its integration also introduces non-negligible system level overhead. For this reason, this study conducts qualitative and quantitative modeling analysis of blockchain-related overhead based on end-to-end training.

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Assuming that only one on-chain transaction needs to be submitted for global knowledge index broadcasting and model state confirmation per round, the end-to-end blockchain delay can be decomposed as follows:

$$T_{bc} = T_{tx} + T_{cons} + T_{exec}. \quad (64)$$

where T_{tx} denotes the transaction propagation delay in the blockchain network, T_{cons} denotes the time required for block consensus, and T_{exec} is the time overhead of smart contract execution and state writing.

(1) **Transaction Propagation and Smart Contract Execution Overhead**. In a permissioned private blockchain environment, the transaction propagation delay T_{tx} is mainly determined by the inter-node communication latency, which is typically on the order of milliseconds and independent of the model parameter dimension d. Meanwhile, since the on-chain contracts in KI-SL are only used for identity authentication, knowledge index hash verification, and state confirmation, and do not participate in any model training or gradient computation, their execution logic is a constant-complexity operation. Thus:

$$T_{exec} = O(1). \quad (65)$$

This overhead does not scale with the growth of training scale or model complexity.

(2) **Consensus Cost and Scalability**. In the case of adopting BFT consensus mechanisms, the consensus communication complexity is typically $O(N_{bc}^2)$, where N_{bc} denotes the number of blockchain nodes participating in the consensus. It is worth noting that in practical IoV deployment architectures, the scale of blockchain consensus nodes is not equivalent to that of vehicular nodes participating in collaborative learning. Typically, consensus nodes consist of a controlled number of infrastructure nodes, such as Road Side Units (RSUs) or edge servers, while vehicular nodes only act as regular clients to submit transaction requests. Therefore, N_{bc} is generally maintained within a relatively stable small scale, which effectively constrains the communication overhead incurred by the consensus process.

More importantly, the blockchain consensus process does not lie on the critical path of model training. The local training, knowledge distillation, and neighbor retrieval processes of KI-SL can all be executed in parallel off-chain, while on-chain operations are only used for result confirmation and index broadcasting. Thus, its latency does not linearly accumulate into each local training iteration.

(3) **Ledger Storage Growth.** As the number of training rounds increases, the storage overhead of the blockchain ledger grows linearly, and its complexity can be expressed as:

$$S_{ledger} = O(T \cdot |\mathcal{U}|). \quad (66)$$

Where T denotes the number of global training rounds, and $|\mathcal{U}|$ is the scale of participating nodes. The on-chain stored content consists of lightweight metadata (e.g., hash indexes, node identifiers, and state digests), rather than raw data, model parameters, or gradient information. Thus, the storage footprint of each individual record is negligible, and the growth of the ledger is primarily driven by the accumulation of historical audit records during long-term system operation.

In practical application deployment, such ledger growth can be effectively managed via widely adopted blockchain data management strategies, including archive nodes, ledger pruning, and hierarchical storage of historical data, thus avoiding sustained pressure on resource-constrained edge devices.

Within the KI-SL framework, since on-chain operations neither involve the transmission of high dimensional model parameters nor participate in model training computation, their time complexity and space overhead exhibit no growth with the model parameter dimension d or the local sample size N. As a result, blockchain related overhead does not dominate the end-to-end training latency. It primarily provides system level support for trusted coordination and auditability, and is fundamentally manifested as low frequency, lightweight, control plane cost decoupled from the model scale, rather than high frequency, high dimensional data plane operations.

In summary, KI-SL delivers pronounced advantages in model performance, computational complexity, communication complexity and security, thereby delivering an efficient and robust solution for data collaboration for the IoV environment under conditions of dynamic heterogeneous data, stringent privacy requirements, and resource constraints.

IV. Experiments and Analysis

A. Experimental Setup

1) Dataset Setup

This experiment was conducted on two real datasets: the Fashion-MNIST dataset [44] and the extended GTSRB dataset (i.e., preprocessed traffic sign data) [45]. The datasets were randomly partitioned and distributed to individual vehicle nodes. Specifically, Fashion-MNIST is a public benchmark dataset for image classification, consisting of 70,000 grayscale clothing images of size 28×28 pixels across 10 categories. The GTSRB dataset was collected by the German Research Center for Traffic Science, with raw data captured by on-board cameras in real road environments; it contains more than 50,000 color traffic sign images from actual driving scenarios across multiple categories. Notably, although the aforementioned datasets cannot fully cover the multimodal data (e.g., LiDAR measurements, vehicular sensor data, and continuous video streams) in real Internet of Vehicles (IoV) systems, as standardized benchmark datasets widely adopted in existing studies, they enable the evaluation of the model performance, communication efficiency, and privacy preservation capability of the proposed method in a controlled experimental environment, while also facilitating fair comparison with existing approaches. To simulate the Non-Independent and Identically Distributed (Non-IID) characteristics among distributed Internet of Vehicles (IoV) nodes, a label-skew-based data partitioning strategy was adopted: the global dataset was first partitioned by category, and then each node was randomly assigned a limited subset of categories, such that different nodes exhibit significant differences in the distribution of their held data categories.

2) Baselines and Standards

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To characterize typical system characteristics of the Internet of Vehicles (IoV) environment at the experimental level, including asynchronous node participation, dynamic availability, and intermittent connectivity, the following modeling assumptions are introduced:(1) **Asynchronous Participation Mechanism**: During each round of global collaboration, only a subset of nodes are randomly activated to perform local training and knowledge upload, while the remaining nodes are in an offline or delayed participation state, which is designed to simulate the asynchronous update behavior of vehicle nodes;(2) **Dynamic Node Join/Exit Mechanism**: Nodes are allowed to dynamically join or exit the collaborative process across different training rounds. Newly joined nodes participate in training solely based on local data and currently available knowledge indexes, without the need for historical model synchronization;(3) **Intermittent Connectivity Mechanism**: Knowledge indexes and gradient updates uploaded by nodes do not require strict round-wise synchronization. KI-SL supports the round-by-round fusion of delayed updates through asynchronous distillation and neighbor matching strategies, thereby avoiding the stagnation of the collaborative training process caused by short-term connection interruptions.

In addition, a convolutional neural network (CNN) based on the LeNet architecture was used to evaluate the training performance of local models. To ensure the validity and stability of the training process, the learning rate was set to 0.01. The standard Stochastic Gradient Descent (SGD) algorithm was adopted for the training process, with a batch size of 8 and the number of local training iterations for participants set to 1. We conducted comparative experiments between KI-SL and the following baseline methods: (1) Swarm Learning (SL) [28]; (2) Federated Knowledge Distillation (FedKD) [21]; (3) Swarm Mutual Learning (SML) [30]; (4) FedAvg [46]; (5) DP-FL [47]; (6) Federated Learning with Quantization Constraints (QFL) [48].

3) Privacy Protection Settings and Evaluation Metrics

The QDP mechanism performs gradient clipping and quantization on the gradients prior to each round of local updates, and injects Laplace noise into the quantized gradients. The gradient clipping threshold C is kept consistent across all experiments, and the quantization granularity is configured with a fixed bit width to ensure the comparability of privacy budgets under different experimental conditions.

For the adversarial privacy protection experiments in Section 4.5, the focus is on evaluating the model's defense capabilities against membership inference attack privacy leakage risks, feature data reconstruction risks, and model stability risks under the scenario with a maximum node count of 120. The specific evaluation objectives include: (1) the defense capability of KI-SL against membership inference attacks; (2) analyzing the feature reconstruction risk caused by the leakage of model update information; (3) evaluating the impact of malicious nodes uploading tampered update information on model stability. The specific evaluation metrics are as follows:

(1) Membership Inference Attack aims to determine whether a specific sample has participated in model training, and its success rate directly reflects the degree of the model's memorization of training data. Based on the attack model proposed by Shokri et al.[49] and the subsequent extension of the statistical discriminant theory of attacks by Yeom et al.[50], the Attack Success Rate (ASR) is adopted as the core evaluation metric, defined as:

$$ASR = \frac{TP+TN}{N}, (0 \leq ASR \leq 1). \tag{67}$$

Where TP (True Positive) denotes the number of data samples correctly identified by the attacker as training members, TN (True Negative) denotes the number of data samples correctly identified by the attacker as non-members, and N denotes the total number of test samples. When ASR is close to 0.5, the attack effect is close to a random guess level, indicating that the model has strong privacy protection capabilities; when $0.6 \leq ASR < 0.7$, the model faces a moderate risk of membership information leakage; when ASR is higher than 0.7, the model has a significant risk of membership privacy leakage.

(2) Gradient Inversion Attack recovers original training samples by analyzing model parameter updates or gradient information. To quantitatively evaluate the attacker's reconstruction capability, this paper constructs a multi-dimensional image quality evaluation system using the **Peak Signal to Noise Ratio (PSNR)** and the **Structural Similarity Index Measure (SSIM)**. The PSNR is the most commonly used pixel-level error metric in the field of image restoration, and has been adopted by Zhu et al.[51] for evaluating the effectiveness of gradient inversion attacks. It is defined as:

$$PSNR = 10\, log_{10} \frac{MAX^2}{MSE}. \tag{68}$$

Where MAX denotes the maximum pixel value of the image (usually 255), and MSE refers to the Mean Squared Error, defined as:

$$MSE = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \left(I(i,j) - \hat{I}(i,j) \right)^2. \tag{69}$$

$I(i,j)$ represents the pixel value of the original image, $\hat{I}(i,j)$ represents the pixel value of the reconstructed image, and H, W denote the height and width of the image, respectively. When PSNR is less than 20 dB, the reconstruction is severely distorted; when PSNR ranges from 20 to 30 dB, the recovery quality is moderate; when PSNR is between 30 and 40 dB, the recovery quality is high; when PSNR exceeds 40 dB, the reconstruction is almost undistorted. A higher PSNR value indicates stronger reconstruction capability of the attacker and thus more severe privacy leakage risk.

(3) The SSIM was proposed by Wang et al.[52], which is designed to measure the degree of semantic similarity between images at the structural level. It is defined as:

$$SSIM(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, (-1 \leq SSIM \leq 1). \tag{70}$$

where μ_x, μ_y denote the mean values of the two images; σ_x^2, σ_y^2 represent their variances; σ_{xy} is the covariance between them; and C_1, C_2 are stability constants introduced to avoid unstable calculation results caused by too small denominator values.

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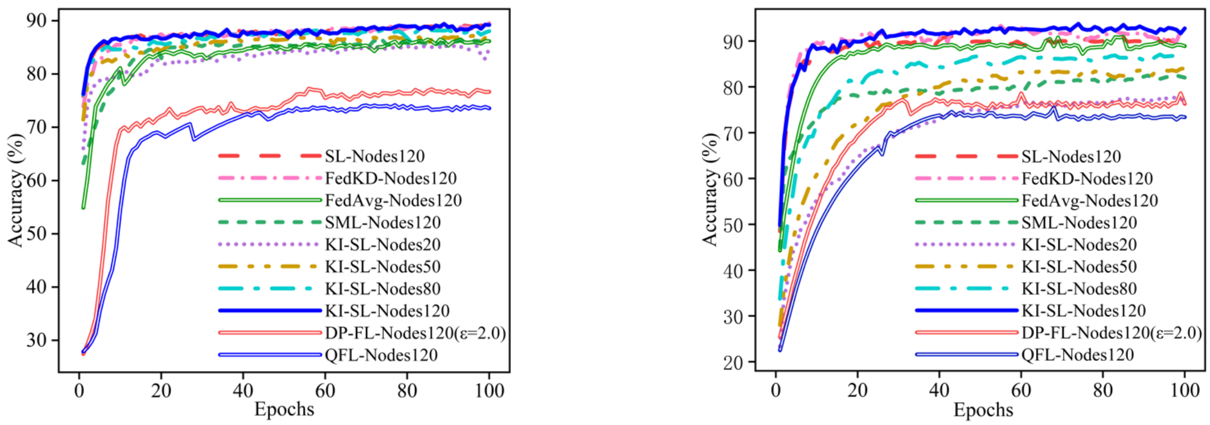
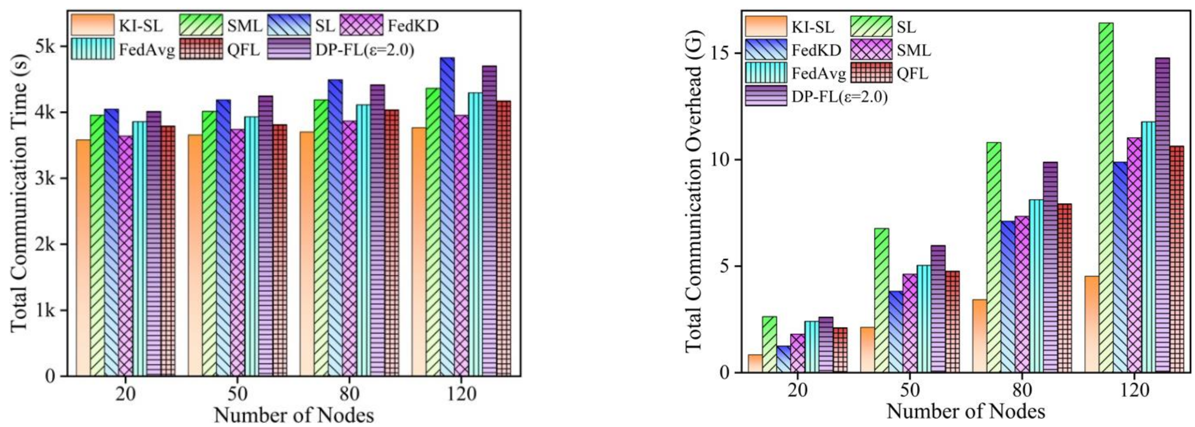


(a) Fashion-MNIST (b) Extended GTSRB

Fig. 4. Comparison of model accuracy under different number of nodes.



(a) Comparison of total communication time (b) Comparison of total communication overhead

Fig. 5. Comparison of communication efficiency under different number of nodes.

When SSIM is close to 1, it indicates that the reconstructed image has a high degree of structural consistency with the original one, implying a severe risk of privacy leakage. When SSIM ranges from 0.6 to 0.8, the reconstructed image retains moderate structural similarity, corresponding to a medium privacy risk. When SSIM is less than 0.5, the structural information of the reconstructed image is severely damaged, which demonstrates that the model has strong privacy protection capabilities.

B. Analysis of Model performance

To evaluate the impact of node scale on model performance, this paper set the maximum node count to 20, 50, 80, and 120 respectively, and set the number of iterations to 100. A comparative analysis of the model performance of various methods under different node scales was conducted, and the experimental results are shown in Fig. 4.

The experimental results are analyzed as follows:

The experimental results demonstrate that KI-SL achieves superior overall performance compared with other baseline methods. On the Fashion-MNIST dataset, as the number of nodes gradually scales up from 20 to 120, both the convergence speed and final accuracy of KI-SL show a consistent upward trend. Specifically, under the 120-node condition, the model accuracy stabilizes at nearly 90%, which is consistently higher than that of other methods. In contrast to DP-FL with differential privacy noise and QFL with quantization constraints, KI-SL avoids the degradation of model accuracy caused by privacy perturbations while ensuring training stability. On the extended GTSRB dataset, KI-SL still maintains a distinct performance advantage. Overall, KI-SL effectively enhances the collaboration efficiency among nodes and verifies its robustness in distributed collaborative learning scenarios.

C. Analysis of Communication efficiency

To evaluate the impact of node count on model communication efficiency, an experiment focusing on communication time and communication overhead was conducted on the Fashion-MNIST dataset, and a comparative analysis of the communication efficiency of various methods under different node counts was performed. The experimental results are shown in Fig. 5.

The experimental results are analyzed as follows:

(1) Fig. 5(a) illustrates the variation of communication time of various methods under different node scales. The results show that as the number of nodes scales up from 20 to 120, the communication time of all methods exhibits an upward trend, while KI-SL maintains consistently lower communication time. At the node count of 20, KI-SL has alr-

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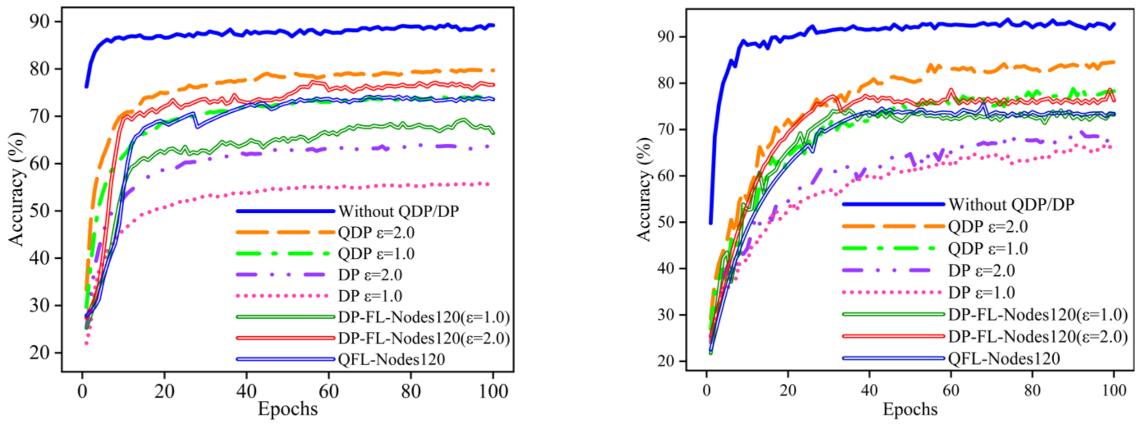

(a) Fashion-MNIST (b) Extended GTSRB

Fig. 6. Comparison of model accuracy of different privacy budgets.

eady shown significant superiority over methods such as SL, FedKD, SML, and FedAvg; as the node scale expands, the communication time of KI-SL still exhibits a relatively gentle growth rate. Under the 120-node condition, the communication time of SL and DP-FL increases most significantly. In addition, DP-FL incurs additional processing overhead due to differential privacy perturbations, resulting in its overall communication time remaining consistently higher than that of KI-SL. Although QFL reduces the amount of data transmitted per round, the cumulative communication costs caused by multi-round interactions still fail to match the efficiency level of KI-SL.

(2) Fig. 5(b) presents the variation of communication overhead of various methods. The results indicate that as the node scale expands, the communication burden of all methods increases significantly, with SL showing the most dramatic growth. In contrast, KI-SL maintains low communication overhead across all node scales and exhibits a slower growth trend. Under the 80-node and 120-node conditions, the communication gap between KI-SL and methods including SL, DP-FL, and FedAvg further widens, demonstrating that KI-SL can effectively reduce redundant parameter exchanges and significantly improve communication efficiency. Overall, KI-SL exhibits distinct communication performance advantages over existing methods in complex collaborative training scenarios.

D. Noise error analysis of quantization differential privacy

To evaluate the impact of different privacy budgets on model performance, the privacy budgets were set to *no noise*, 1.0, and 2.0, respectively, and comparative analyses were conducted between the proposed method and traditional differential privacy (DP), DP-FL, and QFL methods. Among these, the total privacy budget across multiple training rounds was calculated cumulatively in accordance with the standard sequential composition theorem. The experimental results are shown in Fig. 6. The results are analyzed as follows:

Fig. 6(a) and (b) illustrate the accuracy comparison results of various methods under different privacy budgets on the Fashion-MNIST and extended GTSRB datasets, respectively. The results show that after the introduction of differential privacy mechanisms, the accuracy of all methods exhibits a decline, while KI-SL based on Quantized Differential Privacy (QDP) consistently demonstrates significant performance advantages. When the privacy budget ε=2.0, the convergence speed and final accuracy of QDP-KI-SL are significantly superior to those of traditional DP methods. When ε=1.0, even though the overall model accuracy decreases, this method still maintains relatively high accuracy and a more stable convergence trend. Moreover, across the entire privacy budget range, QDP-KI-SL retains consistent performance advantages over DP-FL and QFL. Notably, for the GTSRB dataset, due to its higher task complexity, the performance gap between different methods becomes more pronounced. Based on the combined results of the two groups of experiments, it can be concluded that the QDP mechanism can enhance privacy protection capabilities while maintaining superior model performance, thus effectively mitigating the interference of privacy noise on model training.

E. Analysis of Adversarial Privacy Protection

To verify the effectiveness of the KI-SL method in terms of privacy protection, privacy attack experiments were designed based on adversarial threat models. The comparison results of membership inference attack success rates are presented in TABLE II, the evaluation results of reconstruction attacks (PSNR and SSIM) are listed in TABLE III; and the experimental results of tampering attack robustness are summarized in TABLE IV.

The experimental results are analyzed as follows:

(1) TABLE II presents the comparison results of ASR of different methods under membership inference attacks. The results indicate that traditional methods generally have a high risk of membership privacy leakage. Specifically, the ASR of FedAvg on the two datasets reach 81.4% and 78.9%, respectively, showing a significant training sample memorization effect. In contrast, although SL, SML, and FedKD reduce the ASR to a certain extent, they still pose a moderate risk of privacy leakage. DP-FL further reduces the attack success rate under different privacy budget conditions, but its overall ASR remains higher than the random guess level. By comparison, KI-SL achieves relatively low ASR

under both privacy budget conditions: it decreases to 56.6% and 54.7% when $\varepsilon = 2.0$, and further drops to 49.5% and 47.2% when $\varepsilon = 1.0$, which is lower than the random attack level. This demonstrates that KI-SL can significantly weaken the model's dependence on individual training samples and has distinct advantages in membership privacy protection.

TABLE II
Comparison of Membership Inference Attack Success Rates

Method	ASR (%)	
	Fashion-MNIST	Extended GTSRB
FedAvg	81.4	78.9
SL	75.7	71.3
SML	63.7	61.5
FedKD	65.1	66.8
QFL	77.5	75.2
DP-FL (ε=2.0)	73.2	70.4
DP-FL (ε=1.0)	66.8	65.3
KI-SL (ε=2.0)	56.6	54.7
KI-SL (ε=1.0)	49.5	47.2

TABLE III
Evaluation Results of Reconstruction Attacks

Method	Fashion-MNIST		Extended GTSRB	
	PSNR (dB)	SSIM	PSNR (dB)	SSIM
FedAvg	30.6	0.81	28.7	0.79
SL	23.2	0.77	21.5	0.74
SML	16.9	0.58	15.8	0.57
FedKD	18.4	0.63	17.2	0.61
QFL	24.6	0.79	23.8	0.75
DP-FL (ε=2.0)	22.8	0.75	20.7	0.72
DP-FL (ε=1.0)	21.3	0.72	19.6	0.69
KI-SL (ε=2.0)	14.2	0.51	13.7	0.49
KI-SL (ε=1.0)	13.6	0.47	13.0	0.45

TABLE IV
Comparison of Robustness Against Tampering Attacks

Method	Average Accuracy Degradation (%)
FedAvg	18.2
SL	15.4
SML	12.6
FedKD	14.3
QFL	13.8
DP-FL	13.3
KI-SL	6.2

(2) TABLE III shows the evaluation results of image reconstruction quality under gradient inversion attacks. The results reveal that FedAvg achieves the highest PSNR and SSIM values on the two datasets, indicating that attackers can recover the original training data with high quality. Although SL, FedKD, and SML reduce the reconstruction capability to some extent, they still have an obvious risk of structural recovery. QFL and DP-FL further weaken the attack effect through gradient perturbation, but their PSNR values still remain above approximately 20dB. In contrast, KI-SL exhibits lower PSNR and SSIM indicators under both privacy budget conditions. This suggests that attackers can hardly recover effective image structures, which significantly reduces the risk of training data reconstruction and verifies the outstanding privacy protection capability of KI-SL in resisting gradient inversion attacks.

(3) TABLE IV presents the comparison results of the robustness of various methods under the scenario of model update tampering attacks. The experimental results show that the average accuracy degradation of FedAvg reaches 18.2% under malicious update interference, indicating weak robustness. Although SL, FedKD, and QFL alleviate the impact of attacks to a certain extent, their accuracy still decreases by 12.6% to 15.4%. DP-FL controls the performance degradation at 13.3% through privacy perturbation. In contrast, KI-SL only causes an average accuracy degradation of 6.2%, which is much lower than that of other comparative methods. This indicates that KI-SL can effectively suppress the negative impact of abnormal node updates on the global model and exhibit stronger robustness in complex adversarial environments.

V. Ablation Study

To investigate the impact of different factors on the performance of KI-SL, we conducted an ablation study and designed experiments for key data factors and KI-SL components, respectively. All ablation experiments were carried out on the CIFAR-10 dataset, and the experimental settings in Section V. A are the same as those in Section IV. A. For Section V. B, the Dirichlet distribution ($\boldsymbol{\alpha}$=0.5) was adopted for data partitioning.

A. Ablation Study on Data Factors

To evaluate the impact of key data factors on the performance of the KI-SL model, ablation experiments were designed for three influencing factors, namely data heterogeneity degree, local sample ratio, and number of relevant samples. Comparative analyses were conducted with three baseline methods, including SL, FedKD, and SML, and the experimental results are shown in Fig. 7.

The experimental results are analyzed as follows:

(1) Fig. 7(a) presents the performance comparison results of KI-SL, SL, FedKD, and SML under different degrees of data heterogeneity. The results show that as the data heterogeneity degree increases from 1 to 10, the accuracy of all four methods exhibits a downward trend; however, KI-SL mainta-

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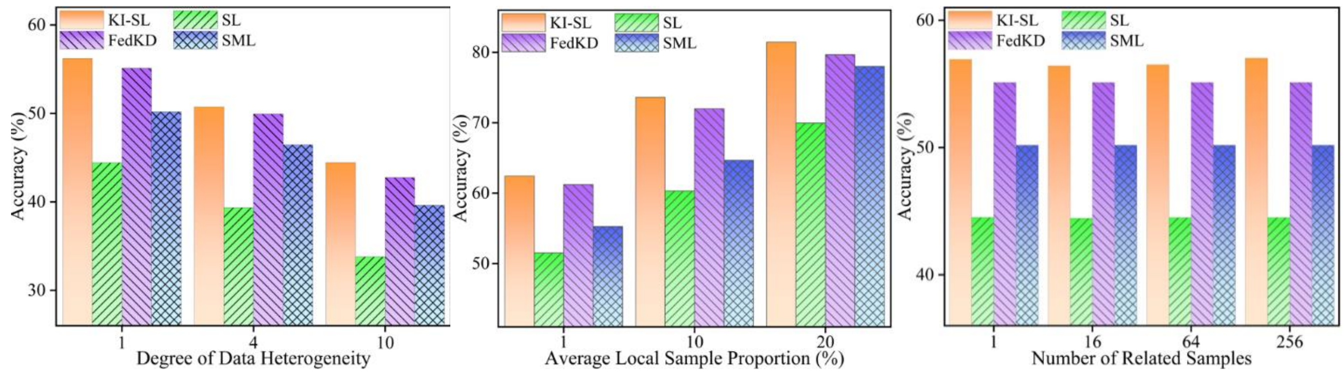

(a) Data heterogeneity (b) The proportion of local samples (c) The number of relevant samples

Fig. 7**.** Comparison of the influence of data factors on model performance.

TABLE V
Comparison of the Impact of Each Component on Performance

Method	Accuracy (%)	Communication Overhead(G)	ASR (%)	PSNR (dB)	SSIM
KI-SL	56.92	4.53	55.21	18.12	0.48
Remove HNSW	53.02	6.58	61.46	21.77	0.61
Remove τ	53.55	5.97	59.92	20.11	0.55
Remove Q	54.26	4.91	57.31	19.24	0.51

TABLE VI
Impact of Adaptive Sampling Parameters

	$\tau = 0$	$\tau = 0.3$	$\tau = 0.5$	$\tau = 0.8$	$\tau = 1$
Accuracy (%)	53.55	56.92	53.23	51.27	50.48

ins a relatively high accuracy across all heterogeneity levels. When the data distribution tends to be balanced (Degree=1), all methods can achieve high accuracy, but KI-SL exhibits the most significant performance advantage. With the intensification of data heterogeneity, although the performance of all methods decreases to a certain extent, the decline rate of KI-SL is relatively gentle, and it still significantly outperforms the three baseline methods. This demonstrates that the KI-SL method possesses strong adaptability and robustness in extremely heterogeneous environments.

(2) Fig. 7(b) presents the accuracy comparison results of the four methods under different local sample ratios. The results show that as the local sample ratio increases from 1% to 20%, the performance of all methods improves as the amount of data increases. Among them, KI-SL exhibits the optimal performance across all ratios; in particular, when the sample ratio is low (1%–10%), its accuracy is significantly higher than that of SL and SML, and slightly higher than that of FedKD. This indicates that KI-SL can effectively address the insufficient training caused by local data scarcity through its efficient knowledge index mechanism. When the sample ratio increases to 20%, KI-SL's accuracy is still higher than that of the three baselines, demonstrating its good adaptability and robustness to changes in data volume.

(3) Fig. 7(c) presents the accuracy comparison results of the four methods under different numbers of relevant samples. The results show that as the number of relevant samples increases from 1 to 256, KI-SL maintains a leading advantage in performance throughout. When the number of relevant samples is extremely small, although all four methods can achieve basic model training, KI-SL achieves higher accuracy compared to the three baseline methods, this reflects its strong robustness under data-sparse conditions. After the number of relevant samples increases, KI-SL's performance remains stable, with no significant fluctuations observed. This indicates that the KI-SL method can achieve relatively stable performance under environments with different scales of relevant samples, effectively addressing the issues of sparse data or unbalanced samples.

B. Ablation Study on KI-SL Components

To analyze the contribution of each module in the proposed KI-SL method to performance improvement, this experiment designed experiments for the HNSW neighbor matching algorithm, adaptive sampling parameter, and prototype optimization module (defined as symbols HNSW, τ and Q) to evaluate the contribution of each component. The performance comparison results are shown in TABLE V, and the results of the impact of adaptive sampling parameters on accuracy are presented in TABLE VI.

The experimental results are analyzed as follows:

(1) TABLE V evaluates the contribution of each core component of KI-SL to the overall performance. The results indicate that the complete KI-SL achieves the optimal balance among accuracy, communication efficiency, and privacy protection. When the HNSW neighbor matching module is

removed, the model accuracy decreases to 53.02%, the communication overhead increases significantly to 6.58G, and both the ASR and image reconstruction quality are obviously improved. This demonstrates that this module plays a core role in improving knowledge matching accuracy, reducing redundant communication, and weakening attack recovery capability.

(2) After removing the adaptive sampling parameter τ, the model accuracy drops to 53.55%, the communication overhead rises to 5.97G, and the privacy attack indicators deteriorate synchronously. This indicates that adaptive sampling can effectively screen high-value knowledge indexes, improve distillation efficiency, and reduce the exposure of sensitive information. TABLE VI further analyzes the impact of the adaptive sampling parameter τ: when $\tau < 0.5$, the model accuracy maintains a high level, while when τ is close to 1, the performance decreases significantly, indicating that overly strong sampling constraints will weaken knowledge interaction.

(3) When the prototype optimization module Q is removed, the model accuracy decreases to 54.26%, the communication overhead increases to 4.91G, the ASR rises to 57.31%, and the PSNR and SSIM increase to 19.24 dB and 0.51, respectively, leading to an increased risk of privacy leakage. This indicates that this module plays an important role in enhancing the stability of knowledge expression and structural consistency.

VI. Conclusion

This study proposes an efficient and secure Knowledge Index based Swarm Learning method (KI-SL) of data collaboration, aiming to address the key challenges of limited communication efficiency and insufficient privacy protection in distributed data collaboration environments such as the Internet of Vehicles (IoV). By designing a knowledge index driven cross-node distillation mechanism, this method achieves efficient knowledge sharing and collaborative model updates among distributed nodes without relying on centralized parameter aggregation, thereby significantly improving the learning stability in heterogeneous data scenarios. Meanwhile, the designed quantized differential privacy (QDP) mechanism provides privacy protection for the processes of knowledge transmission and gradient update while controlling communication overhead, effectively reducing the risk of sensitive information leakage. In multi-node collaboration scenarios, KI-SL significantly reduces system communication costs and enhances the collaboration between models, with its core advantages reflected in communication efficiency, privacy controllability, and good adaptability to heterogeneous nodes and asynchronous updates.

Compared with existing swarm learning and federated distillation frameworks, KI-SL breaks through the limitations of existing methods from parameter synchronization to global aggregation and from output distillation to centralized coordination, providing a new implementation path for large-scale decentralized collaborative learning. Future research will focus on the following two directions: (1) Dynamic grouping mechanism: Combining state attributes such as dynamic node computing power, communication status, and changes in data distribution, research adaptive dynamic grouping and intra-group reconstruction strategies to further improve the stability and scalability of swarm learning in highly dynamic environments; (2) Secure aggregation algorithm: On the premise of maintaining the decentralized characteristics, introduce more efficient secure aggregation and verifiable computing mechanisms to enhance the defense capability against malicious nodes

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Haiyan Kang is a senior member of the China Computer Federation (No. E200028533M), ACM Membership (No.9495204), and member of privacy protection committee of China Confidentiality Association. He received his Ph.D. in computer application technology from Beijing Institute of Technology, China in 2005. He has published more than 130 papers. His research interest fields include information system security and privacy preserving, etc.

Chong Zhang is currently a master's candidate with School of Computer Science, Beijing Information Science and Technology University of China. His research interests include federated learning, data sharing, and privacy preserving.